Cheat Sheet:the Hadoop Ecosystem

Package/Method	Description	Code Example
bin/hadoop	All Hadoop commands are invoked by the bin/hadoop script. Running the Hadoop script without any arguments prints the description for all commands.	Running Hadoop script without arguments: `bin/hadoop`

cat	Reads each file parameter in sequence and writes it to standard output. If you do not specify a file name, the cat command reads from standard input. You can also specify a file name of - (dash) for standard input.	Create two sample files.
		`echo "This is file 1" > file1.txt`
		`echo "This is file 2" > file2.txt`
		Use the cat command to read and display the contents of both files
		`cat file1.txt file2.txt`
		Sample output (Contents of file1.txt and file2.txt):
		`This is file 1`
		`This is file 2`

cd	Used to move efficiently from the existing working directory to different directories on your system.	Basic syntax of cd command:
		`cd [options]... [directory]`
		Example 1: Change directory location to "folder1"
		`cd /usr/local/folder1`
		Example 2: Get back to the previous working directory
		`cd -`
		Example 3: Move up one level from the present working directory tree
		`cd ..`

create table	Used to create a new table in a database	Create a new database (if not already created).
		`CREATE DATABASE your_database;`
		Use the newly created database.
		`USE your_database;`
		Create a new table named "employees" in Hive.
		`CREATE TABLE employees (`
		`    id INT,`
		`    first_name STRING,`
		`    last_name STRING,`
		`    email STRING,`
		`    hire_date DATE`
		`)`
		`ROW FORMAT DELIMITED`
		`FIELDS TERMINATED BY ','`
		`STORED AS TEXTFILE;`
		Show the list of tables in the database.
		`SHOW TABLES;`
		Sample Output (List of Tables):
		`OK`
		`employees`

curl

A command-line tool (pronounced "curl") that allows data to be exchanged between a device and a server through a terminal. The user specifies the server URL, the location where they want to send the request, and the data they want to send to the server URL using this command-line interface (CLI).

Example 1: Sending a GET request and displaying the response

Send a GET request to a server and display the response.

```
curl https://www.example.com
```

In this example, we use the curl command to send a GET request to https://www.example.com and display the HTML response from the server.

Example 2: Sending data to a server using POST Request:

Send a POST request with data to a server and display the response.

```
curl -X POST -d "name=John&age=30" https://www.example.com/api
```

In this example, we use the curl command to send a POST request to https://www.example.com/api with data `name=John&age=30` and display the JSON response from the server.

docker exec

Runs a new command in a running container. It only runs when the container's primary process is running, and it is not restarted if the container is restarted.

Running a command in a running Docker container:

Run a new command inside a running Docker container.

```
docker exec -it container_name_or_id ls /app
```

Sample Output (List of files in the '/app' Directory inside the container):

```
file1.txt
file2.txt
subdirectory
```

In this example:

- `docker exec` is used to run a new command (`ls /app`) inside a running Docker container.
- `-it` enables an interactive terminal session, which allows you to see the output of the command.
- `container_name_or_id` is the name or ID of the running Docker container you want to execute the command in.
- `ls /app` is the command that lists the files and directories in the '/app' directory inside the container.

docker-compose

Compose is a tool for defining and running multi-container Docker applications. It uses the YAML file to configure the services and enables us to create and start all the services from just one configuration file.

Starting Docker containers using docker-compose:

Suppose you have a `docker-compose.yml` file like this:

```
version: '3'
    services:
        web:
            image: nginx:latest
            ports:
              - "80:80"
        db:
            image: postgres:latest
            environment:
            POSTGRES_PASSWORD: example_password
```

You can use `docker-compose` to start the services defined in the docker-compose.yml file as follows:

Navigate to the directory containing the docker-compose.yml file.

```
cd /path/to/your/docker-compose-project
```

Start the Docker containers defined in the docker-compose.yml file

```
docker-compose up
```

docker pull	You can download Docker images from the internet.	`docker pull [OPTIONS] IMAGE_NAME[:TAG]`
docker run	It runs a command in a new container, getting the image and starting the container if needed.	`docker run [OPTIONS] IMAGE [COMMAND] [ARG...]`
git clone	You can create a copy of a specific repository or branch within a repository.	`git clone REPOSITORY_URL [DESTINATION_DIRECTORY]`

hdfs dfs	Apache Hadoop hadoop fs or hdfs dfs are file system commands to interact with HDFS. These commands are very similar to Unix commands. Hadoop provides two types of commands to interact with the file system: hadoop fs or hdfs dfs. The major difference is that Hadoop commands are supported with multiple file systems like S3, Azure, and many more.	Example-1:
		Listing files and directories in HDFS:
		List files and directories in the root directory of HDFS.
		`hdfs dfs -ls /`
		Example-2: In this example, we use the `hdfs dfs -ls` command to list files and directories in the root directory of HDFS.
		`hdfs dfs -ls /`
		Sample output:
		`drwxr-xr-x   - hdfs hduser          0 2023-09-13 10:00 /user`
		`drwxrwxrwx   - hdfs hduser          0 2023-09-13 10:05 /tmp`
		`drwxrwxrwx   - mapred hduser        0 2023-09-13 10:10 /mapred`
		Create a new directory named "mydata" in HDFS.
		`hdfs dfs -mkdir /user/your_username/mydata`

hdfs dfs -cat	Display the contents for a file.	Display the contents of a file in HDFS. `hdfs dfs -cat /path/to/file.txt`
hdfs dfs -mkdir	Creates a directory named path in HDFS	Create a directory in HDFS. `hdfs dfs -mkdir /user/username/mydirectory`
hdfs dfs -put	Upload a file or folder from the local disk to HDFS.	Upload a file from the local file system to HDFS. `hdfs dfs -put localfile.txt /user/username/hdfsfile.txt`

LOAD DATA INPATH	Hive provides the functionality to load precreated table entities either from the local file system or from HDFS. This command is used to load data into the hive table.	Load data from HDFS into a Hive table.
		`LOAD DATA INPATH '/user/username/hdfsfile.txt' INTO TABLE`
		`mytable;`

ls	Writes to standard output the contents of each specified Directory parameter or the name of each specified file parameter, along with any other information you ask for with the flags. If you do not specify a file or directory parameter, the ls command displays the contents of the current directory.	Basic command syntax
		`ls [options] [file/directory]`
		Example 1: Sorts the file names displayed in the order of last modification time. 'r' is for displaying in reverse order
		`ls -lt`
		`ls -ltr`
		Example 2: Displays hidden files
		`ls -a`

mkdir	Used to create one or more directories specified by the Directory parameter. Each new directory contains the standard entries dot (.) and dot dot (..). You can specify the permissions for the new directories with the -m Mode flag.	Create a new directory named "myfolder." `mkdir myfolder`
SELECT * FROM	Lists all the rows from the table to check if the data has been loaded from the file.	Select all rows from a table. `SELECT * FROM tablename;`

show tables	Used to see all the tables in the database that have been selected.	Show all tables in the selected database. `SHOW TABLES;`
tar	Looks for archives on the default device (usually tape) unless you specify another device. When writing to an archive, the tar command uses a temporary file (the /tmp/tar* file) and maintains in memory a table of files with several links.	Create a tar archive of a directory. `tar -cvf archive.tar /path/to/directory`

wget	Stands for web get. The wget is a free, noninteractive file downloader command. Noninteractive means it can work in the background when the user is not logged in.	Basic syntax of the wget command; commonly used options are `[-V], [-h], [-b], [-e], [-o], [-a], [-q]`
		`wget [options]... [URL]...`
		Example 1: Specifies to download file.txt over HTTP website URL into the working directory.
		`wget http://example.com/file.txt`
		Example 2: Specifies to download the archive.zip over the HTTP website URL in the background and returns you to the command prompt in the interim.
		`wget -b http://www.example.org/files/archive.zip`

Apache	This open-source HTTP server implements current HTTP standards to be highly secure, easily configurable, and highly extendible. The Apache Software License by the Apache Software Foundation builds and distributes it.

Apache ZooKeeper	A centralized service for maintaining configuration information to maintain healthy links between nodes. It provides synchronization across distributed applications. It also tracks server failure and network partitions by triggering an error message and then repairing the failed nodes.
Big data	Data sets whose type or size supersedes the ability of traditional relational databases to manage, capture, and process the data with low latency. Big data characteristics include high volume, velocity, and variety.
Big data analytics	Uses advanced analytic techniques against large, diverse big data sets, including structured, semi-structured, and unstructured data, from varied sources and sizes, from terabytes to zettabytes.
Block	Minimum amount of data written or read, and also offers fault tolerance. The default block size can be 64 or 128 MB, depending on the user's system configuration. Each file stored need not take up the storage of the preconfigured block size.
Clusters	These servers are managed and participate in workload management. They allow enterprise applications to supersede the throughput achieved with a single application server.
Command-line interface (CLI)	Used to enter commands that enable users to manage the system.

Commodity hardware	Consists of low-cost workstations or desktop computers that are IBM-compatible and run multiple operating systems such as Microsoft Windows, Linux, and DOS without additional adaptations or software.
Data ingestion	The first stage of big data processing. It is a process of importing and loading data into IBM® WatsonX.data. You can use the Ingestion jobs tab from the Data manager page to load data securely and easily into WatsonX.data console.
Data sets	Created by extracting data from packages or data modules. They gather a customized collection of items that you use frequently. As users update their data set, dashboards and stories are also updated.

Distributed computing	A system or machine with multiple components on different machines. Each component has its own job, but the components communicate with each other to run as one system for the end user.
Driver	Receives query statements submitted through the command line and sends the query to the compiler after initiating a session.
Executor	Executes tasks after the optimizer has split the tasks.
Extended Hadoop Ecosystem	Consists of libraries or software packages commonly used with or installed on top of the Hadoop core.
Fault tolerance	A system is fault-tolerant if it can continue performing despite parts failing. Fault tolerance helps to make your remote-boot infrastructure more robust. In the case of OS deployment servers, the whole system is fault-tolerant if the servers back up each other.
File system	An all-comprehensive directory structure with a root (/) directory and other directories and files under a logical volume. The complete information about the file system centralized in the /etc/filesystems file.

Impala	A scalable system that allows nontechnical users to search for and access the data in Hadoop.
InputSplits	Created by the logical division of data. They serve as an input to a single Mapper job.
JDBC client	Component in the Hive client allows Java-based applications to connect to Hive.

Low latency data access	A type of data access allowing minimal delays, not noticeable to humans, between an input processed and corresponding output offering real-time characteristics. It is crucial for internet connections using trading, online gaming, and Voice over IP.

Meta store	Stores the metadata, the data, and information about each table, such as the location and schema. In turn, the meta store, file system, and job client communicate with Hive storage and computing to perform the following: Metadata information from tables store in some databases and query results, and data loaded from the tables store in a Hadoop cluster on HDFS.
Node	A single independent system for storing and processing big data. HDFS follows the primary/secondary concept.
ODBC (Open Database Connectivity) Client	Component in the Hive client, which allows applications based on the ODBC protocol to connect to Hive.
Optimizer	Performs transformations on the execution and splits the tasks to help speed up and improve efficiency.
Parallel computing	Workload for each job is distributed across several processors on one or more computers, called compute nodes.
Parser	A program that interprets the physical bit stream of an incoming message and creates an internal logical representation of the message in a tree structure. The parser also regenerates a bit stream from the internal message tree representation for an outgoing message.
Partitioning	This implies dividing the table into parts depending on the values of a specific column, such as date or city.

Primary node	Also known as the name node, it regulates client file access and maintains, manages, and assigns tasks to the secondary node. The architecture is such that per cluster, there is one name node and multiple data nodes, the secondary nodes.
Rack	The collection of about forty to fifty data nodes using the same network switch.
Rack awareness	When performing operations such as read and write, the name node maximizes performance by choosing the data nodes closest to themselves. Developers can select data nodes on the same rack or

	nearby racks. It reduces network traffic and improve cluster performance. The name node keeps the rack ID information to achieve rack awareness.

Replication	The process of creating a copy of the data block. It is performed by rack awareness as well. It is done by ensuring data node replicas are in different racks. So, if a rack is down, users can obtain the data from another rack.
Replication factor	Defined as the number of times you make a copy of the data block. Users can set the number of copies they want, depending on their configuration.
Schema	It is a collection of named objects. It provides a way to group those objects logically. A schema is also a name qualifier; it provides a way to use the same natural name for several objects and prevent ambiguous references.

Read	In this operation, the client will request the primary node to acquire the location of the data nodes containing blocks. The client will read files closest to the data nodes.

Write	In this operation, the Name node ensures that the file does not exist. If the file exists, the client gets an IO Exception message. If the file does not exist, the client is given access to start writing files.
Yet Another Resource Negotiator (YARN)	Prepares Hadoop for batch, stream, interactive, and graph processing.